

Team Discussion Brief and Meeting Agenda

Cross-Generator Generalization in Deepfake Detection | IE7374: Generative AI
Prepared from our background, dataset, and methodology research; all figures are traceable to the cited papers.

Summary

We train standard CNN (convolutional neural network) deepfake detectors and measure how much their accuracy drops on fakes produced by a manipulation method they were **not trained on**. That drop — not a single headline accuracy — is the project's main result. As a one-line summary: a detector that scores 96% on the method it was trained on but 51% on an unseen method is not really a 96% detector, and reporting that second number honestly is our contribution.

Key terms

So the report and the meeting use the same vocabulary:

CNN	Convolutional Neural Network — the standard image-classification model we use.
GAN	Generative Adversarial Network — a common method for generating deepfakes.
AUC / AUROC	Area Under the ROC Curve — our main accuracy measure. 100 = perfect, 50 = random guessing. AUC and AUROC mean the same thing.
FF++	FaceForensics++ — our primary dataset of real and manipulated face videos.
DFDC	DeepFake Detection Challenge — a second dataset, used only as an optional test.
LOMO	Leave-One-Manipulation-Out — train on 3 of the 4 methods, test on the 4th.
In-distribution	Tested on a method the model was trained on (the easier case).
Cross-generator	Tested on a method the model was not trained on (the harder case).

Suggested meeting agenda (about 45 minutes)

Time	Item
5 min	Align on the one-sentence summary and why this is a Generative-AI project
10 min	Review the key research findings (Sections 1–2)
10 min	Settle the open decisions (Section 5) — the main goal of the meeting
10 min	Split the 8 training runs and confirm role ownership (Section 6)
10 min	Review risks and how we mitigate them (Section 7)

1. Why this matters

- Fewer than **1% of people** can reliably identify a deepfake (Zupan, 2025).
- Deepfakes are already used for fraud, harassment, and misinformation; the U.S. Government Accountability Office (GAO) and UNESCO identify them as a threat to information integrity (GAO 2024; Naffi 2025).

- Detection lags generation: each new generation method produces a new pattern of artifacts that existing detectors have not seen — for example, the shift from GAN-based to diffusion-based methods.

This remains a Generative-AI project even though we do not train a generator: the content we detect is the output of generative models, and the central challenge is the distribution shift from one generation method to another.

2. Key finding from the literature

Detectors that are strong on a method they were trained on (in-distribution) lose much of their accuracy on methods they were not trained on (cross-generator). Several studies report the same pattern:

Setting	In-distribution	Unseen method / dataset	Source
FF++ → Celeb-DF (cross-dataset)	99.7 AUC	48.2 AUC (below random)	Celeb-DF, CVPR 2020
FF++ leave-one-out (Xception)	~96 AUC	77.9 avg; 51.2 on FaceSwap	LipForensics, CVPR 2021
GAN-trained → diffusion images	97.4 AUC	78.6 (detects 26% of fakes)	Ricker et al., 2024
FF++ → DFDC (cross-dataset)	0.96 AUC	~0.71 AUC	DeepfakeBench, 2023

The clearest single result is the 48.2 AUC: an AUC below 50 means the detector performs worse than random guessing on data from a new source.

3. Dataset facts (FaceForensics++)

- **1,000 real videos + 4,000 fakes = 5,000 total.** The same 1,000 originals are manipulated by four different methods.
- **The four methods form a clean 2×2**, which is the basis of our transfer hypotheses (shown below).

	Face-swap (changes identity)	Reenactment (keeps identity)
Learning-based	DeepFakes	NeuralTextures
Graphics-based	FaceSwap	Face2Face

- **We use the c23 (high-quality) version** — about 20 GB total — which fits our storage and preprocessing budget. The raw version is roughly 1 TB and is not needed.
- **The official 720/140/140 split is identity-disjoint by design:** the same person never appears in both training and test data, which avoids a common cause of inflated results (the model recognizing the person rather than the manipulation).
- The data is fake-heavy (1 real : 4 fake); we balance it per method (1,000 vs 1,000).

FaceForensics++ is preferable to a single large dataset because it provides four cleanly separated manipulation methods built on the same faces. This lets us isolate the effect of an unseen manipulation method from the effect of an unseen person, which is exactly what the project measures.

4. Method

- **Two models:** EfficientNet and XceptionNet, both pretrained on ImageNet and fine-tuned (not trained from scratch). XceptionNet is the standard FF++ baseline; EfficientNet is the backbone used by the DFDC competition winner.
- **Protocol — Leave-One-Manipulation-Out (LOMO):** train on 3 of the 4 methods, test on the 4th. Rotating which method is held out gives 4 setups × 2 models = **8 training runs**.
- **Main deliverable:** a transfer matrix — trained-on-method-X versus tested-on-method-Y for every pair — with the in-distribution versus cross-generator gap as the headline result.
- **Feasibility:** about 80,000–160,000 cached face images; the 8 runs take days, not weeks, on a single 12 GB GPU. The bottleneck is preprocessing (cropping faces once), not GPU time.

The 8 runs are independent of one another, so team members can each own one or two and the full set finishes quickly — which also makes the workload easy to divide evenly.

5. What we expect to find

- **In-distribution: about 0.96 AUC** — the expected, routine case.
- **Held-out (unseen): about 0.55–0.85 AUC** — a 15-to-40-point drop. This gap is the result; a large gap is a meaningful finding, not a failure.
- **NeuralTextures** is expected to be the hardest method to detect when held out, because it is the subtlest manipulation.
- **FaceSwap** is expected to be the weakest training source, because its graphics-based artifacts transfer poorly to learning-based methods.
- **DeepFakes and Face2Face** are expected to transfer best to each other, because their artifacts are most similar.

Caveat to state for a fair comparison: most published numbers train on one method, whereas we train on three. The added variety should make our gap somewhat smaller than the published single-method figures, and we will note this so the comparison is fair.

6. Decisions to settle today

#	Decision	Recommendation
1	Run both EfficientNet and XceptionNet, or one?	Both — allows an architecture comparison
2	Pretrained or trained from scratch?	Pretrained (faster; course-recommended)
3	Is the DFDC cross-dataset test in scope?	Stretch goal only (preprocessing time)
4	Who owns which of the 8 training runs?	Split 2 models × 4 folds across the team
5	Confirm or adjust the role assignments (Sec. 6)	Sagar's area is the open one
6	Official FF++ request form or Kaggle mirror?	Form = clean license; mirror = faster

Note: decisions 1–3 were pre-selected for the draft proposal; this is the team's opportunity to confirm or change them.

7. Roles (confirm or trade)

Member	Area	Responsibilities
Dominic Rivas	Coordination, evaluation, analysis	Protocol, transfer matrix, figures

Jonathan Regalado	Exploratory data analysis	Class balance, identity-overlap audit
Lyxelis R. Navarro	Preprocessing pipeline	Face cropping and cached feature store
Obinna Okonkwo	Model implementation	EfficientNet/Xception, training, checkpoints
Sagar Ayare	Documentation, reporting, DFDC stretch	Literature review, report editing, cross-dataset test

Roles are a starting point and may be traded. Each member co-writes the report section tied to their area and reviews one other section.

8. Risks and how we mitigate them

- **Identity leakage** (the same face in training and test) inflates results. Mitigation: the FF++ official split already prevents this; Jonathan audits to confirm.
- **Re-decoding video every epoch** wastes time. Mitigation: crop faces once, cache to disk, and have everyone train from the cache.
- **Using the held-out method during tuning** would invalidate the result. Mitigation: keep the held-out method's files separate until final evaluation.
- **Comparing our LOMO numbers to single-method papers** is not a like-for-like comparison. Mitigation: state the caveat from Section 5.
- **The full DFDC set is about 470 GB.** Mitigation: use the Preview subset (about 5,000 clips) only, and only if time allows.

9. Key phrases for the report and presentation

- “Cross-generator generalization must be measured, not assumed.”
- “A detector that scores 96% on a known method and 51% on an unseen one is the result of interest — we report the gap, not the peak.”
- “Each new generation method is a distribution shift the detector was never trained on.”
- “We report an honest characterization of where standard CNN detectors break, rather than a single best-case number.”

Sources: figures are documented in .planning/research/ (BACKGROUND.md, DATASETS.md, METHODOLOGY_CNN.md). Primary papers: FaceForensics++ (Rössler et al., 2019), LipForensics (Haliassos et al., 2021), Celeb-DF (Li et al., 2020), DeepfakeBench (Yan et al., 2023), Ricker et al. (2024).